Nguyen Viet Anh

Data Engineer | Applied Machine Learning | Ho Chi Minh City
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Summary

Data Engineer at OSAS via AI-MED and HCMUT Computer Science graduate. Build Oracle ETL and Python applications; evaluate tabular ML, wearable-health models and OCR with reproducible experiments and explicit validation protocols.

Education

Ho Chi Minh City University of Technology (HCMUT) 2022 – 2026
Bachelor of Computer Science, Information Systems – Graduated

Experience

Data Engineer – OSAS Joint Stock Company via AI-MED Jun 2025 – Present

- Develop Oracle warehouse loads from three source systems using PL/SQL and ODI 12c, including incremental loading and SCD Type 2 dimensions. Configured GoldenGate CDC for a separate public-sector warehouse.
- Reconciled source and warehouse records; corrected sales-channel classifications across approximately 68,000 rows and a policy-counting error that returned 82 policies instead of one for a customer.
- Built a Vietnamese financial-statement assistant with Python, FastAPI, and Ollama: section summaries, rule-based figure extraction, financial ratios, and source-page references. Implemented SQL/vector retrieval and an ingestion queue.

Projects

Adult Income – Classical ML and PyTorch Oct 2026
Coursework extension; official UCI data | GitHub

- Built train-only preprocessing and validation-based model/threshold selection; removed feature duplicates and train/test overlap before separate test evaluation.
- LightGBM achieved 86.73% accuracy and 0.9276 ROC-AUC on 16,249 test rows. PyTorch MLP accuracy was 84.09–84.50% across three seeds with validation-based early stopping.

HealthMate – Wearable Health Monitoring Sep 2025 – May 2026
Graduation thesis; team of three | Backend | Flutter app

- Designed the Flutter interface and helped evaluate stress models using 869 labeled windows from 15 WESAD participants. Held out one participant at a time to assess performance on people outside the training set.
- Evaluated prescription OCR on 66 manually labeled samples. PaddleOCR with OpenCV and extraction rules achieved medication-name micro-F1 of 0.46, compared with 0.24 for PaddleOCR-VL; used the comparison to guide the app's OCR choice.

Vietnam Stock Data Pipeline 2025 – 2026
Personal project; streaming and warehouse | GitHub

- Connected VNStock, Kafka, PySpark Structured Streaming, TimescaleDB and Snowflake; used incremental sync, persisted watermarks, Bronze/Silver/DW layers and Streams & Tasks.
- Built Streamlit/Plotly dashboards and Docker Compose services with Nginx, ZooKeeper, Kafka/Spark monitoring, pgAdmin and Linux/systemd operations.

Technical Skills

Languages & ML: Python, SQL, PL/SQL, Dart; NumPy, Pandas, scikit-learn, CatBoost, LightGBM, PyTorch, imbalanced-learn.
Streaming & ETL: Kafka/kafka-python, Spark/PySpark, VNStock, ODI, GoldenGate.
Databases: Oracle, PostgreSQL/TimescaleDB, pgvector, Snowflake (Streams & Tasks).
Apps & BI: FastAPI, Flutter, Streamlit, Plotly; psycopg2, JDBC, Snowflake Connector.
Operations: Git, Linux/Bash/systemd, Docker Compose, Nginx, ZooKeeper, Kafka UI, Spark UI/History Server, pgAdmin.

Certifications

Python Project for Data Engineering – IBM/Coursera, 2026.
Relational Database Administration (DBA) – IBM/Coursera, 2026. | All certificates